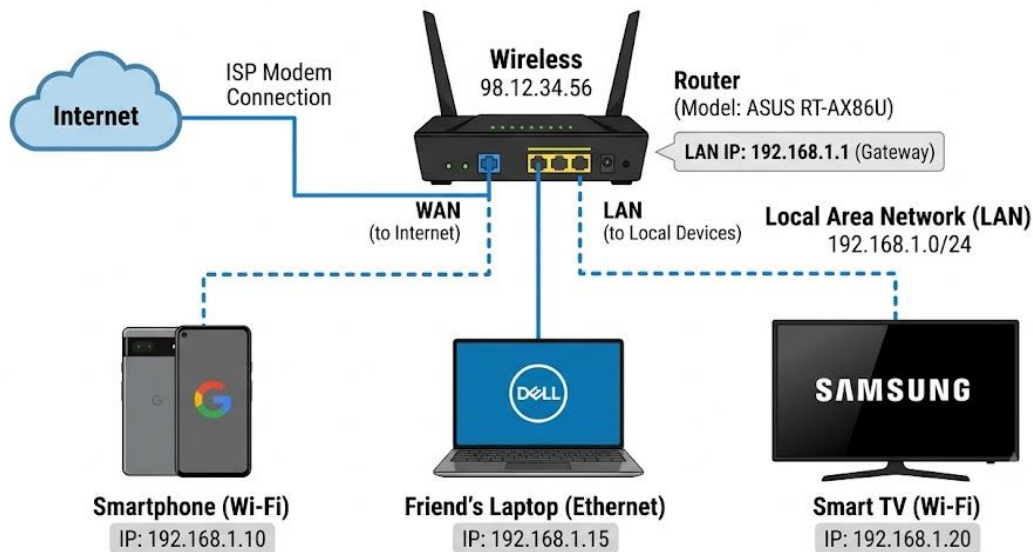


IP routing and routing protocols

Task – 1 Labelling the IP address through router.

HOME NETWORK DIAGRAM: DEVICE CONNECTIONS & IP ADDRESSES



Task – 2 How packet will travel to laptop from computer.

- Network Setup & IP Addressing
- Your Laptop (Host A): 192.168.1.50/24 | Default Gateway: 192.168.1.1
- Local Router (Router 1):
 - LAN (eth0): 192.168.1.1/24
 - WAN (eth1): 10.0.0.1/30
- Remote Router (Router 2):
 - WAN (eth0): 10.0.0.2/30
 - LAN (eth1): 172.16.0.1/24
- Friend's Laptop (Host B): 172.16.0.100/24 | Default Gateway: 172.16.0.1
- Step 1: Host A Determines Local vs. Remote Destination. Host A wants to send a packet to 172.16.0.100. It performs a bitwise AND operation on the target IP using

its subnet mask (255.255.255.0). Since 172.16.0.0 does not match its local subnet (192.168.1.0), Host A checks its local routing table:

Destination Network	Subnet Mask	Gateway	Interface
192.168.1.0	255.255.255.0	0.0.0.0	eth0
0.0.0.0 (Default)	0.0.0.0	192.168.1.1	eth0

- Decision: The longest prefix match is the default gateway (0.0.0.0/0). Host A encapsulates the IP packet into an Ethernet frame targeted to Router 1's MAC address and sends it out.
- Step 2: Hop 1 — Router 1 Lookup and Forwarding. Router 1 receives the frame, removes the Ethernet header, verifies the IP checksum, and decrements the IP TTL (Time to Live) by 1. It examines the packet's destination IP (172.16.0.100) against its routing table:

Destination Network	Subnet Mask	Gateway	Interface
192.168.1.0	255.255.255.0	0.0.0.0	eth0
10.0.0.0	255.255.255.252	0.0.0.0	eth1
172.16.0.0	255.255.255.0	10.0.0.2	eth1

- Decision: The IP 172.16.0.100 matches entry 172.16.0.0/24. The next-hop gateway is 10.0.0.2 via interface eth1. Router 1 re-encapsulates the IP packet into a new layer 2 frame and sends it across the WAN link to 10.0.0.2.
- Step 3: Hop 2 — Router 2 Local Delivery. Router 2 receives the frame on eth0, strips the layer 2 header, verifies the IP header, and decrements the TTL again. It checks its routing table for 172.16.0.100:

Destination Network	Subnet Mask	Gateway	Interface
10.0.0.0	255.255.255.252	0.0.0.0	eth0
172.16.0.0	255.255.255.0	0.0.0.0	eth1

- Decision: The destination falls within its directly connected subnet (172.16.0.0/24) on interface eth1. Because gateway is 0.0.0.0, Router 2 uses ARP (Address Resolution Protocol) on its eth1 interface to find Host B's MAC address associated with 172.16.0.100. Router 2 encapsulates the IP packet into a final Ethernet frame and transmits it directly to Host B.

Task – 3 Different subnet, address, subnet mask.

- Delivery Hub Network Configuration
- Branch 1 (Mumbai Hub): 10.10.0.0/16 | Interface: eth1
- Branch 2 (Delhi Hub): 10.20.0.0/16 | Interface: eth2
- Branch 3 (Bengaluru Hub): 10.30.0.0/16 | Interface: eth3
- Core Transit Network: 192.168.254.0/24 (Connects central routers via eth0)
- Core Router Central Routing Table

Destination Network	Subnet Mask	Next Hop (Gateway)	Interface	Route Type / Description
10.10.0.0	255.255.0.0	10.10.0.1	eth1	Directly Connected (Mumbai POS & Order Processing)
10.20.0.0	255.255.0.0	10.20.0.1	eth2	Directly Connected (Delhi Delivery Fleet Dispatch)
10.30.0.0	255.255.0.0	10.30.0.1	eth3	Directly Connected (Bengaluru Live Tracking Services)
192.168.254.0	255.255.255.0	0.0.0.0	eth0	Directly Connected (Inter-Hub Transit Backbone)

0.0.0.0	0.0.0.0	192.168.254.254	eth0	Static Default Gateway (External Cloud Services & APIs)
---------	---------	-----------------	------	--

Task – 4 Comparison of Static Routing and Dynamic Routing

- Static Routing :- Routes are manually configured by a network administrator and remain fixed unless changed by hand.
- Pros:
 - Low Overhead: Zero CPU, RAM, or bandwidth overhead since routers do not exchange routing updates.
 - High Security: Admins have total control over traffic flow; packets cannot be rerouted by spoofed protocol updates.
 - Predictable Pathing: Traffic always takes the exact specified route, making troubleshooting straightforward.
- Cons:
 - No Automatic Failover: If a link drops, traffic is silently lost until an admin manually changes the table.
 - Poor Scalability: Management becomes unfeasible as the network grows to hundreds of subnets or hubs.
 - Human Error Risk: Manual updates on large networks easily lead to configuration mistakes and routing loops.
- Dynamic Routing Protocols (RIP, OSPF, BGP). :- Routers automatically discover routes, build routing tables, and adapt to network topology changes in real time.
- Pros:
 - Automatic Failover: Instantly reroutes traffic around dead links or congested network nodes without manual intervention.
 - High Scalability: New subnets and devices join the network automatically once added to the protocol domain.
 - Optimal Path Calculation: Protocols like OSPF dynamically calculate the fastest, lowest-cost path based on bandwidth and link state.
- Cons:

- Resource Consumption: Consumes router CPU, memory, and link bandwidth to process routing algorithms (e.g., Dijkstra's algorithm) and exchange periodic updates.
- Security Vulnerabilities: Susceptible to routing table poisoning or rogue router insertions if authentication isn't configured.
- Initial Complexity: Requires complex initial setup and deep protocol knowledge to tune timers and metric costs properly.

Task – 5 OSPF routing between two routers.

- Here is a sample Cisco IOS configuration for setting up OSPF (Open Shortest Path First) between two routers, along with a line-by-line breakdown of what each command actually does.
- Router 1 (R1) Configuration

```

- Plaintext
- Router1# configure terminal
Router1(config)# interface GigabitEthernet0/0
Router1(config-if)# ip address 10.0.0.1 255.255.255.252
Router1(config-if)# no shutdown
Router1(config-if)# exit
Router1(config)# router ospf 1
Router1(config-router)# router-id 1.1.1.1
Router1(config-router)# network 10.0.0.0 0.0.0.3 area 0
Router1(config-router)# network 192.168.1.0 0.0.0.255 area 0

```

- Line-by-Line Breakdown :-configure terminal: Enters global configuration mode so you can make system-wide changes to the router. Interface GigabitEthernet0/0: Selects the physical network port connected to Router 2 to start configuring its IP settings.
- ip address 10.0.0.1 255.255.255.252: Assigns the IP address and a /30 subnet mask to the link connecting the two routers.
- no shutdown: Powers on the interface (Cisco ports are disabled by default).
- exit: Leaves the interface settings to go back to global configuration mode.
- router ospf 1: Starts the OSPF routing process on the router. The 1 is a local Process ID used by the router to keep track of this instance.
- router-id 1.1.1.1: Gives the router a unique 32-bit ID within the OSPF network. This prevents ties and identifies who originated specific routing updates.

- `network 10.0.0.0 0.0.0.3 area 0`: Tells OSPF to enable the protocol on the WAN link interface (10.0.0.1). The 0.0.0.3 is a wildcard mask (the inverse of 255.255.255.252), and area 0 assigns it to the primary OSPF backbone area.
- `network 192.168.1.0 0.0.0.255 area 0`: Enables OSPF on the local LAN interface (192.168.1.0/24) so Router 1 can advertise its local network to Router.

-

- Router 2 (R2) Configuration

- Plaintext

```
Router2# configure terminal
Router2(config)# interface GigabitEthernet0/0
Router2(config-if)# ip address 10.0.0.2 255.255.255.252
Router2(config-if)# no shutdown
Router2(config-if)# exit
Router2(config)# router ospf 1
Router2(config-router)# router-id 2.2.2.2
Router2(config-router)# network 10.0.0.0 0.0.0.3 area 0
Router2(config-router)# network 172.16.0.0 0.0.0.255 area 0
```

- Line-by-Line Breakdown :- `interface GigabitEthernet0/0 & ip address 10.0.0.2 255.255.255.252`: Configures the opposite end of the point-to-point link facing Router 1 with the corresponding IP on the same /30 subnet.
- `router ospf 1`: Initializes OSPF locally on Router 2.
- `router-id 2.2.2.2`: Sets Router 2's unique identifier.
- `network 10.0.0.0 0.0.0.3 area 0`: Activates OSPF on the link shared with Router 1. Once both routers exchange OSPF Hello packets on this matching area (area 0) and subnet, they form an OSPF neighbor adjacency.
- `network 172.16.0.0 0.0.0.255 area 0`: Instructs Router 2 to advertise its local LAN (172.16.0.0/24) to Router 1 across the newly established OSPF neighbor link.

-